

6(a)	At $V = 12 \text{ V}$, $I = 2.5 \text{ A}$ $\Rightarrow \frac{I}{V} = 0.208$ At $V = 6 \text{ V}$, $I = 1.25 \text{ A}$ $\Rightarrow \frac{I}{V} = 0.208$ At $V = 9.6 \text{ V}$, $I = 2.0 \text{ A}$ $\Rightarrow \frac{I}{V} = 0.208$ Since ratio of I to V is constant, I is proportional to V.	[1] show at least 2 values of ratio of $\frac{I}{V}$ or $\frac{V}{I}$. [1] conclusion
6(b)(i)	Resistance of resistor $X = \frac{12}{2.5} = 4.8 \Omega$	[1] answer
6(b)(ii)1.	Current in wire AB = $\frac{9.0}{4.0 + 5.0} = 1.0 \text{ A}$	[1] answer
6(b)(ii)2.	Current in resistor X = $\frac{9.0}{4.8 + 2.7} = 1.2 \text{ A}$	[1] answer
6(b)(iii)	Resistance of wire AC = $\frac{0.70}{1.0} \times 4.0 = 2.8 \Omega$ Potential difference across AC = $\frac{2.8}{4.0 + 5.0} \times 9.0 = 2.8 \text{ V}$ Potential difference across AD = $\frac{4.8}{4.8 + 2.7} \times 9.0 = 5.76 \text{ V}$ Potential difference between C and D = $5.76 - 2.8 = 2.96 \text{ V}$ = 3.0 V	[1] value of V_{AC} [1] value of V_{AD} [1] answer
6(b)(iv)	With internal resistance, the <u>potential difference across the power supply would be less than 9.0 V.</u> The potential difference across AC and AD would also be less, and hence the <u>answer in (iii) would be less.</u>	[1]
7(a)	$V = 40 \text{ V}$ (40)	